AIM: Association Rule Mining

- Implement the Association Rule Mining algorithm (e.g., Apriori) to find
- frequent itemsets.
- Generate association rules from the frequent itemsets and calculate their
- support and confidence.
- Interpret and analyze the discovered association rules.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules


df = pd.read_csv("C:/Users/hrane/Downloads/online food delivery_107.csv")
df.columns = df.columns.str.strip()
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)


columns_to_drop = [
    "Age",
    "Family size",
    "latitude",
    "longitude",
    "Pin code",
    "Unnamed: 13",
]
X_categorical = df.drop(columns=columns_to_drop, errors="ignore")
X_encoded = pd.get_dummies(X_categorical, dtype=bool)


frequent_itemsets = apriori(X_encoded, min_support=0.3, use_colnames=True)

rules = association_rules(
    frequent_itemsets, metric="confidence", min_threshold=0.7
)


rules = rules[
    (rules["antecedents"].apply(lambda x: len(x)) <= 1)
    & (rules["consequents"].apply(lambda x: len(x)) <= 1)
]


rules["rule_name"] = rules.apply(
    lambda row: f"{list(row['antecedents'])} -> {list(row['consequents'])}",
    axis=1,
)
```

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```
top_rules = rules.sort_values(by="lift", ascending=False).head(10)

plt.figure(figsize=(12, 6))

bars = plt.barh(
    top_rules["rule_name"],
    top_rules["lift"],
    color="#990033",
    edgecolor="black",
    height=0.6,
)


plt.title("Top 10 Association Rules by Lift Strength", fontsize=14, pad=15)
plt.xlabel("Lift Value (Rule Strength Multiplier)", fontsize=12)
plt.ylabel("Association Rules (Antecedent -> Consequent)", fontsize=12)
plt.grid(axis="x", linestyle="--", alpha=0.5)
plt.gca().invert_yaxis()


for bar in bars:
    width = bar.get_width()
    plt.text(
        width + 0.02,
        bar.get_y() + bar.get_height() / 2,
        f"{width:.2f}",
        ha="left",
        va="center",
        fontweight="bold",
    )

plt.tight_layout()
plt.show()
```

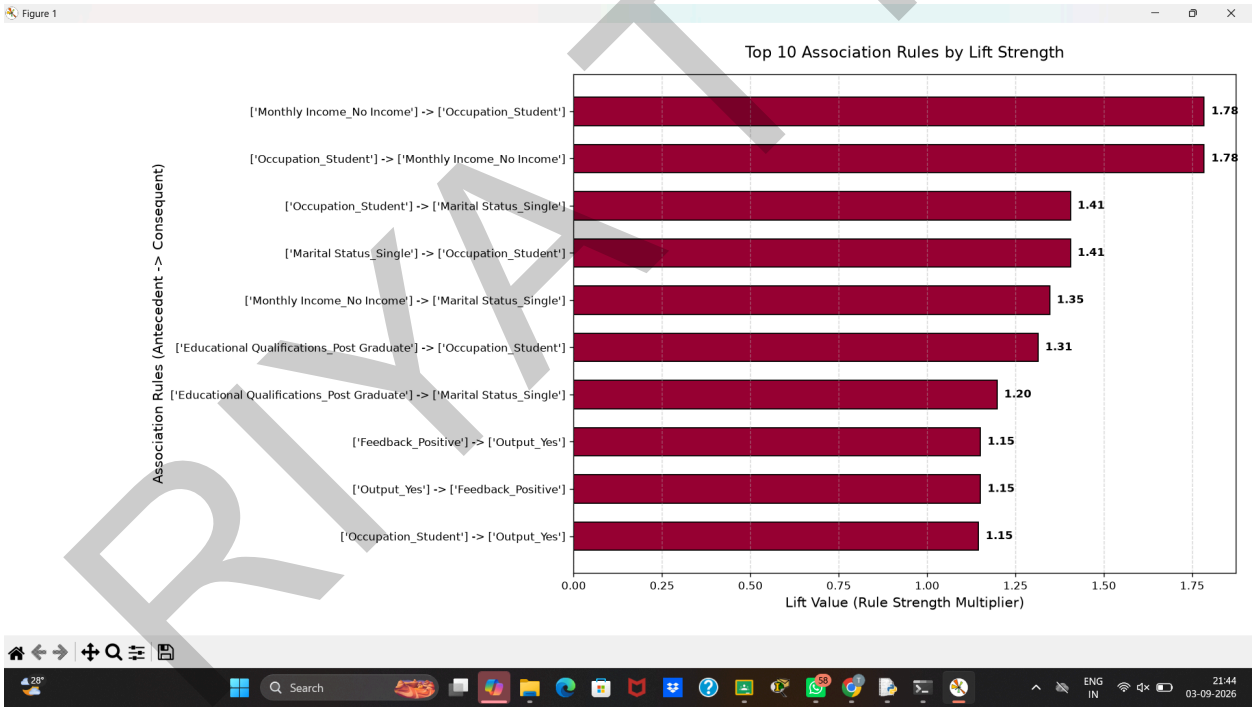
OUTPUT:

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```
================== RESTART: C:\Users\hrane\Documents\AI_R.py ==============
============================================================
Total Frequent Itemsets Discovered: 135
============================================================
    support                                           itemsets
0  0.427835                      frozenset({Gender_Female})
1  0.572165                        frozenset({Gender_Male})
2  0.690722                 frozenset({Marital Status_Single})
3  0.304124                   frozenset({Occupation_Employee})
4  0.533505                    frozenset({Occupation_Student})
5  0.481959               frozenset({Monthly Income_No Income})
6  0.456186   frozenset({Educational Qualifications_Graduate})
7  0.448454  frozenset({Educational Qualifications_Post Gra...
8  0.376289                frozenset({Customer Type_Frequent})
9  0.561856                 frozenset({Customer Type_Regular})

============================================================
--- TOP DISCOVERED ASSOCIATION RULES (Sorted by Lift) ---
============================================================
                                           antecedents  ...      lift
613  [Feedback_Positive, Occupation_Student, Unname...  ...  2.157013
612  [Feedback_Positive, Occupation_Student, Output...  ...  2.157013
610  [Output_Yes, Monthly Income_No Income, Marital...  ...  2.157013
611  [Monthly Income_No Income, Marital Status_Sing...  ...  2.157013
461  [Monthly Income_No Income, Marital Status_Sing...  ...  2.108696

[5 rows x 5 columns]
```



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